GABRIEL ORTIZ

Available Jan - Jun 2027 | 956-505-1628 | gabriel@gabrielortiz.io | github.com/gabrielortiz00 | linkedin.com/in/gabrielortiz00

EDUCATION

Northeastern University, Boston, MA — Expected May 2029
Candidate for Bachelor of Science
Major: Computer Science | Dean's List, Spring 2026
Relevant Coursework: Object-Oriented Design, Linear Algebra, Discrete Structures, Program Design & Implementation 1

SKILLS

- *Programming Languages:* Python, Java, JavaScript, SQL, HTML/CSS
- *Frameworks/Libraries:* NumPy, Pandas, Matplotlib, asyncio, JavaFX, Mapbox GL JS
- *Tools:* Git, REST APIs, pytest, QGIS
- *Additional:* FAA Private Pilot Certificate (PPL), Spanish (Native), German (B2), DJ (10+ years, 3,000+ attendees)

TECHNICAL EXPERIENCE

AerospaceNU, *Software Engineer*, Boston, MA — Jul 2026 - Present
- Extended the async data acquisition pipeline for ATLAS, a Python CLI that pulls satellite imagery and climate data from 20+ sources (Sentinel-2, Landsat, MODIS, NASA FLDAS)
- Fixed silent incomplete or misleading results in 4 ATLAS data clients (GOES, Himawari, Maxar, GIBS) by adding shared S3 pagination, antimeridian-aware bounding-box filtering, and event filtering before Maxar's result cap
- Designing mode-transition logic and electronics stack for Mudskipper, a multi-modal air/ground/water drone

PROJECTS

AirViz | JavaScript, Mapbox GL JS, Vite, QGIS | github.com/gabrielortiz00/airviz — Apr 2026 - Present
- Developed an interactive 3D airspace visualizer for student pilots that renders FAA Class B, C and D airspace for 13 terminal areas as stacked volumes, instead of flat sectional charts
- Converted FAA NASR airspace data to GeoJSON in QGIS and split it into per-airport files covering 40+ airspace volumes with published floor and ceiling altitudes

Cost Accumulation Software | Java, JavaFX | github.com/gabrielortiz00/byp-cost-calculator — Sep 2024 - Jan 2025
- Built a Java desktop application for the costing team at BYP, a Mexican paint-applicator manufacturer, replacing Excel formulas with an internal tool that calculates product cost from shipping, taxes, container limits, and import fees
- Implemented user authentication, product database persistence, live exchange rate API integration, and cost breakdown charts
- Created a pricing calculator that sets a price from a target profit margin, or works backward from a desired price to the resulting margin

LEADERSHIP & ADDITIONAL EXPERIENCE

STEMinds Club, *Founder,* Monterrey, Mexico — Sep 2023 - May 2025
- Founded STEMinds Club and organized a state-level science fair with 300+ participants; raised $2,500 in sponsorships
- Led twice-monthly sessions on computer fundamentals and online safety for groups of 20 kids in underserved communities

MHAS - Social Action for Improvement of Communities, *Finance Director*, Monterrey, Mexico — Oct 2023 - May 2025
- Coordinated community improvement trips to low-income areas in Mexico with 40+ students and corporate professionals
- Directed the re-painting of 3 under-resourced public schools and 15 homes for people with mental health conditions

Kumon Institute of Education, *Teaching Assistant*, Monterrey, Mexico — Mar 2024 - Oct 2024
- Tracked progress and maintained records for 30 students, and communicated results to parents and instructors
- Guided students in foundational and advanced math and English